

Matheus Lima de Araujo

PhD Candidate

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RESEARCH EXPERIENCE

I study uncertainty in biodiversity big data and the mechanisms influencing global biodiversity patterns, with a focus on developing quantitative approaches in macroecology. Although my research now focuses more on general questions than on specific taxa, I have a background in plant ecology, particularly epiphytes. I have also collaborated across diverse fields within community ecology, including landscape ecology and conservation biology.

Here are some of my keywords: Big-data, community ecology, functional traits, knowledge shortfalls, macroecology, macroevolution, phylogenetics, software development, uncertainty.

EDUCATION

2023 – Present: PhD in Ecology and Evolution

Federal University of Goias (UFG), Goiania, Brazil.

Thesis: From gaps to insights: Quantifying biological data uncertainty and revisiting biodiversity in environmental space.

Supervisor: Dr. Thiago F. Rangel

2025 – 2026: Internship abroad as a PhD student

Swiss Federal Research Institute for Forest, Snow and Landscape (WSL), Zurich, Switzerland.

Funding: Brazilian Federal Agency for Support and Evaluation of Graduate Education - CAPES

Supervisor: Dr. Catherine H. Graham

2021 – 2023: MSc in Ecology and Evolution

Federal University of Goias (UFG), Goiania, Brazil.

Dissertation: Interpreting deep evolutionary patterns through recent ecological processes.

Supervisor: Dr. Thiago F. Rangel

2016 – 2021: Bachelor in Biological Sciences

Federal University of Alfenas (UNIFAL), Minas Gerais, Brazil.

Bachelor's thesis: Biases in sample collection shape our knowledge of vascular epiphytes.

Supervisor: Dr. Flavio Nunes Ramos

TEACHING EXPERIENCE

2022 – 2022 – Teaching internship in **Statistics** at the Universidade Federal de Goias. Supervisor: Dr. Thiago Rangel. Duration: 64 hours.

2021 – 2021 – Teaching internship in **Statistics** at the Universidade Federal de Goias. Supervisor: Dr. Thiago Rangel. Duration: 64 hours

2020 – 2020 – Teaching in **Introduction to R in Environmental Sciences** at the Universidade Federal de Alfenas. Supervisor: Prof. Dr. Flavio Nunes Ramos. Duration: 50 hours.

2020 – 2020 – Volunteer undergraduate teaching assistant in **Evolutionary Ecology** at the Universidade Federal de Alfenas. Supervisor: Prof. Dr. Flavio Nunes Ramos. Duration: 60 hours.

2018 – 2018 – Volunteer undergraduate teaching assistant in **Invertebrate Zoology II** at the Universidade Federal de Alfenas. Supervisor: Prof. Dr. Humberto Fonseca Mendes. Duration: 112 hours.

2017 – 2018 – Volunteer undergraduate teaching assistant in **Invertebrate Zoology I** at the Universidade Federal de Alfenas. Supervisor: Prof. Dr. Humberto Fonseca Mendes. Duration: 90 hours.

SUPERVISION EXPERIENCE

2026 – Present: Supervising undergraduate student Cecilia Correa e Silva (Federal University of Goias – BR) on a proposal for Scientific Initiation Scholarship project: “Investigating body size patterns for lizards and snakes in environmental space”.

2024 – 2025: Supervising undergraduate student Beatryz Nascimento Ibiapina (Federal University of Goias – BR) on Scientific Initiation Scholarship project: “Challenges in the quality of global distribution data for the tiger shark (*Galeocerdo cuvier*): a comparative analysis between GBIF and IUCN”.

2023 – 2023: Co-supervising undergraduate student Gustavo Nunes Freitas (Federal University of Goiás – BR, PI: Dr. Andre Luis Regolin) on the bachelor's thesis: "Landscape simplification reduces the probability of puma occurrence in agroecosystems of the central Brazilian Cerrado".

FUNDING AND AWARDS

2025 – 2026: CAPES Doctoral "Sandwich" Fellowship (competitive award), Brazil. Funded 9-month research stay at Swiss Federal Research Institute for Forest, Snow and Landscape (WSL), Zurich, Switzerland.

2023 – Present: PhD Scholarship (competitive program-wise award), CAPES, Brazil. Awarded after a selection based on written exam in Ecology and Evolution plus CV evaluation.

2021 – 2023: Master's Scholarship (competitive program-wise award), CAPES, Brazil. Awarded after a selection based on written exam in Ecology and Evolution plus CV evaluation.

2020 – 2021: CNPq Scientific Initiation Scholarship (competitive university-wide award), Brazil — Funded undergraduate research at Universidade Federal de Alfenas, MG, BR.

2019 – 2020: CNPq Scientific Initiation Scholarship (competitive university-wide award), Brazil — Funded undergraduate research at Universidade Federal de Alfenas, MG, BR.

2018 – 2019: CNPq Scientific Initiation Scholarship (competitive university-wide award), Brazil — Funded undergraduate research at Universidade Federal de Alfenas, MG, BR.

SCIENTIFIC PUBLICATIONS (chronologically)

12. Nakamura, G., Stabile, B. H. M., Frateles, L. E. F., **Araujo, M. L.**, Neuhaus, E. B., Marinho, M. M. F., De Souza Leite, M., Richter, A., Ding, L., Da Silva Freitas, T. M., Soares, B., Da Graça, W. J., Moura, M. R., & Diniz-Filho, J. a. F. (2025). The hidden biodiversity knowledge split in biological collections. *Proceedings of the Royal Society B Biological Sciences*, 292(2058), 20251045. <https://doi.org/10.1098/rspb.2025.1045>

11. Coelho, M. T. P., Barreto, E., Diniz-Filho, J. a. F., Rangel, T. F., Bohdalková, E., Storch, D., Jang, Y., Pontarp, M., Cassemiro, F., **Araujo, M. L.**, Hull, P. M., & Graham, C. H. (2025). Consistent energy-diversity relationships in terrestrial vertebrates. *Science*, 389(6755), 53–57. <https://doi.org/10.1126/science.adu2590>

10. **Araujo, M. L.**, Coelho, M. T. P., Gotelli, N. J., Colwell, R. K., Rangel, T. F., & Dambros, C. S. (2025). In situ speciation of Dispersal-Limited taxa leads to strong phylogenetic community clustering. *Journal of Biogeography*, 52(9). <https://doi.org/10.1111/jbi.70005>

9. Teles, W. S., Regolin, A. L., De Andrade-Silva, B. E., Maldonado, A., Junior, Vilela, R. D. V., Borges-Almeida, K., Pereira, M. J., **Araujo, M. L.**, & Collevatti, R. G. (2025). Host diversity mediates the influence of landscape structure on parasite communities in Cerrado agricultural landscapes. *Annals of the New York Academy of Sciences*, 1545(1), 105–115. <https://doi.org/10.1111/nyas.15292>
8. Graham, C. H., **Araujo, M. L.**, Barreto, E., Dambros, C. S., Diniz-Filho, J. a. F., Zimmermann, N. E., Rangel, T. F., & Coelho, M. T. P. (2025). Biodiversity patterns redefined in environmental space. *Ecology Letters*, 28(2), e70008. <https://doi.org/10.1111/ele.70008>
7. **Araujo, M. L.**, Coelho, M. T. P., Cassemiro, F. a. S., & Rangel, T. F. (2024). Global Distribution of Mammalian Cradles and Museums is Driven by Past Climate Dynamics and Present Water–Energy Balance. *Global Ecology and Biogeography*, 33(12). <https://doi.org/10.1111/geb.13914>
6. **Araujo, M. L.**, Ferreira, L. G. S. S., Nakamura, G., Coelho, M. T. P., & Rangel, T. F. (2024). ‘treesliceR’: a package for slicing phylogenies and inferring phylogenetic patterns over evolutionary time. *Ecography*, 2025(1). <https://doi.org/10.1111/ecog.07364>
5. Pereira, F. W., **Araujo, M. L.**, Brum, F. T., Melo, G. a. R., Moura, M. O., & Gonçalves, R. B. (2024). Bees travelling south: Climate-induced range shifts and suitable habitat losses in south-eastern neotropics. *Journal of Biogeography*, 51(11), 2259–2273. <https://doi.org/10.1111/jbi.14989>
4. Ramos, F. N., **Araujo, M. L.**, Da Silva Assis, I., Basilio, G. A., Barbosa, D. E. F., Correa, T. A., Delgado, C. N., Furtado, S. G., Marquez, G., Soares, J. S., Yañez, A., Silva, S. M., Neto, L. M., & Elias, J. P. C. (2024). Vascular epiphytes of the South America Dry Diagonal: characterising their occurrence and richness in this neglected region. *New Zealand Journal of Botany*, 63(4), 822–839. <https://doi.org/10.1080/0028825x.2024.2387188>
3. Carneiro, M. S., Ribeiro, B. R., Campos, C. C. F., De Souza Almeida, H., **Araujo, M. L.**, Santos, F. a. M. D., & Ramos, F. N. (2023). Small forest fragments and their importance for conservation of tree communities in Atlantic Semideciduous Forest. *Iheringia Série Botânica*, 78, e2023019. <https://doi.org/10.21826/2446-82312023v78e2023019>
2. **Araujo, M. L.**, Quaresma, A. C., & Ramos, F. N. (2022). GBIF information is not enough: national database improves the inventory completeness of Amazonian epiphytes. *Biodiversity and Conservation*, 31(11), 2797–2815. <https://doi.org/10.1007/s10531-022-02458-x>
1. **Araujo, M. L.**, & Ramos, F. N. (2021). Targeting the survey efforts: Gaps and biases in epiphyte sampling at a biodiversity hotspot. *Forest Ecology and Management*, 498, 119544. <https://doi.org/10.1016/j.foreco.2021.119544>

MANUSCRIPTS UNDER PEER REVIEW

1. Elias, M. A. M., **Araujo, M. L.**, Gawryszewski, F. M., Stevens, M., Lemes, P., & Maciel, N. M. Beyond warning colours: Climate change is the dominant force shaping poison frog distribution. *Under review* in Journal for Nature Conservation.
2. **Araujo, M. L.**, Diniz-Filho, J. A. F., & Rangel, T. F. Global patterns of speciation rates are governed by ecological limits and niche use dynamics. *Under review* in Ecology.
3. Pereira, F. W., **Araujo, M. L.**, Lepeco, A., Marques, B. F., Bampi, H., Jardim, L., Vareira, L., Carvalheiro, L., Rangel, T. F., & Diniz Filho, J. A. F. The missing branches of the tree of life: addressing global Darwinian shortfalls and their drivers. *Under review* in Journal of Biogeography.
4. **Araujo, M. L.**, Diniz-Filho, J. A. F., Pereira, F. W., Ramos, F. N., Coelho, M. T. P., Graham, C. H., & Rangel, T. F. Species-rich tropical regions show high uncertainty in geographical range and environmental niche estimates. *Under review* in Proceedings of the Royal Society B: Biological Sciences.
5. Ibiapina, B. N., Vareira, L., Bampi, H., Rangel, T. F., & **Araujo, M. L.** Distribution mismatches and hidden sampling dynamics: Insights from the tiger-shark (*Galeocerdo cuvier*). *Under review* in Biological Conservation.

MANUSCRIPTS IN PREPARATION

1. **Araujo, M. L.**, Diniz-Filho, J. A. F., Tagliacollo, V. A., Cassemiro, F., Albert, J. S., Rangel, T. F. PAM: A tool for integrating missing species into phylogenetic trees and mapping uncertainty across space and time. *In prep.*
2. **Araujo, M. L.**, Barreto, E., Coelho, M. T. P., Diniz-Filho, J. A. F., Graham, C. H., Olalla-Tárraga, M. A., Rangel, T. F. Bergmann's rule is supported for terrestrial tetrapods in environmental space. *In prep.*
3. **Araujo, M. L.**, Graham, C. H., Zimmermann, N. E., Rangel, T. F. A mid-domain-effect in niche geometry for terrestrial tetrapods. *In prep.*

TECHNICAL PRODUCTION

1. **Araujo, M.**, Ferreira, L., Nakamura, G., Coelho, M., & Rangel, T. (2025). **_treesliceR: To Slice Phylogenetic Trees and Infer Evolutionary Patterns Over Time_. R package version 1.1.0**, <<https://CRAN.R-project.org/package=treesliceR>>.

PROFESSIONAL ACADEMIC SERVICES

Reviewer for the journals: *Austral Ecology* (2026–present), *Biodiversity Informatics* (2026–present), *Ecography* (2026–present), *Global Ecology and Biogeography*

(2025–present), *Nature Communications* (2025–present), *Biotropica* (2025–present), *Diversity and Distributions* (2024–present), *Journal of Biogeography* (2023–present).

CONFERENCE PARTICIPATION & PRESENTATION

2026 - Celebrating Biogeography Research 2026. *A mid-domain rule of environmental niche conservatism in tetrapods* (Oral presentation). University of Bayreuth, Bayreuth, Germany.

2026 - The International Biogeography Society (IBS) 12th Biennial Conference. *Bergmann's rule is supported for terrestrial vertebrates in environmental space* (Poster). Aarhus University, Aarhus, Denmark.

2024 - 1st Brazilian Congress of Evolutionary Biology. *Global distribution of mammalian cradles and museums is driven by past climate dynamics and present water–energy balance* (Poster). Federal University of Paraná (Poster), Curitiba, Paraná, Brazil.

2024 - 4th RABECO - Ecologia Plural: Intersectionality and Transdisciplinarity in Solving Environmental Problems. *treesliceR: a package for slicing phylogenies and inferring phylogenetic patterns over evolutionary time* (Poster). Federal University of Bahia (UFBA), Salvador, Bahia, Brazil.

2021 - 7th Integrated Symposium UNIFAL: Transversality, Science, and Affection. *GBIF information is not enough: national database improves the inventory completeness of Amazonian epiphytes* (Oral presentation). Federal University of Alfenas (UNIFAL), Alfenas, Minas Gerais, Brazil.

2020 - 6th Integrated Symposium UNIFAL - University, Science, and Society. *Geographical Gaps and Biases in the Quality of Digital Data on Atlantic Forest Epiphytes* (Oral presentation). Federal University of Alfenas (UNIFAL), Alfenas, Minas Gerais, Brazil.

2019 - 5th Integrated Symposium UNIFAL. *Effect of Landscape Configuration on Tree Species Richness* (Oral presentation). Federal University of Alfenas (UNIFAL), Alfenas, Minas Gerais, Brazil.

2019 - 14th Open Congress for Biology Students (XIV CAEB). State University of Campinas (UNICAMP), Campinas, São Paulo, Brazil.

2019 - 14th Brazilian Ecology Congress (CEB). São Lourenço, Minas Gerais, Brazil.

2017 - 28th Ecology Study Week. São Paulo State University (UNESP), Rio Claro, São Paulo, Brazil.

PERSONAL SKILLS

Programming languages: R and C++;

Languages: Portuguese (native) & English (fluent).

REFERENCES

Dr. Flavio N. Ramos

Federal University of Alfenas (UNIFAL), Alfenas, Minas Gerais, Brazil.

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Dr. Thiago F. Rangel

Federal University of Goias (UFG), Goiania, Goias, Brazil.

Contact information: thiago.rangel@ufg.br

Dr. Catherine H. Graham

Swiss Federal Research Institute WSL, Birmensdorf, Switzerland.

Contact information: catherine.graham@wsl.ch